

City of Detroit 2022 - 2025 Sales

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28 July 2026

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Introduction

This report evaluates the metrics of a linear regression model created from citywide datasets of sales data from the years 2022 - 2024 and 2023 - 2025. Overlapping sales IDs were removed to include only unique sales, leaving a dataset containing single family residential sales from 2022 - 2025. After removing outliers in the initial data prep, a total of 19756 sales were used for model development.

Custom functions were written to automate variable creation from initial datasets, calculate assessing coefficients (PRD, COD, PRB), variance factors, and graphs.

```
## <environment: 0x000001f1c2277a18>
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Functions Used: modfunctions_notes.R + modelfunctions.R

Model I

Table 1: LN_SPPSF model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic
19748	19723	24	19747	25	0.53473	0.16335	0.16233	0.53439	0.43164	160.4483	0	31345

Table 2: LN_SPPSF model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.9562	222.6419	0.0178	0.0000	3.9214	3.9911
single_family1story	0.0471	4.5878	0.0103	0.0000	0.0270	0.0672
smallest	0.1351	9.0619	0.0149	0.0000	0.1059	0.1644
small	0.0560	5.2285	0.0107	0.0000	0.0350	0.0770
two_plus_baths	0.1648	7.2661	0.0227	0.0000	0.1203	0.2093
powder_room	0.0737	5.9026	0.0125	0.0000	0.0492	0.0982
good	0.3441	11.6741	0.0295	0.0000	0.2863	0.4019
fair	-0.2073	-21.7951	0.0095	0.0000	-0.2259	-0.1886
poor	-0.3591	-16.5107	0.0217	0.0000	-0.4017	-0.3164

term	estimate	statistic	std_error	p_value	conf_low	conf_high
very_poor_unsound	-0.3344	-7.3688	0.0454	0.0000	-0.4234	-0.2455
qabove_avg	0.4437	15.6394	0.0284	0.0000	0.3881	0.4993
qbelow_avg	-0.0352	-3.6021	0.0098	0.0003	-0.0543	-0.0160
yb_pre_1929	-0.1854	-15.5461	0.0119	0.0000	-0.2087	-0.1620
yb_1946_to_1960	0.0763	8.2472	0.0093	0.0000	0.0582	0.0945
ln_garagespaces	0.1059	10.7062	0.0099	0.0000	0.0865	0.1252
fireplace	0.1006	10.7522	0.0094	0.0000	0.0823	0.1190
hot_water	0.0720	4.8426	0.0149	0.0000	0.0429	0.1012
elec_baseboard	-0.3185	-2.2181	0.1436	0.0266	-0.6000	-0.0371
wall	-0.1238	-2.5772	0.0481	0.0100	-0.2180	-0.0297
central_air	0.0926	8.6261	0.0107	0.0000	0.0716	0.1137
months_1to9	-0.0102	-4.8751	0.0021	0.0000	-0.0142	-0.0061
months_10to16	0.0188	8.5730	0.0022	0.0000	0.0145	0.0231
months_17to24	0.0100	10.4898	0.0010	0.0000	0.0081	0.0119
near_downtown_1mi	1.3215	4.9202	0.2686	0.0000	0.7950	1.8479
near_downtown_5mi	0.0420	2.5977	0.0162	0.0094	0.0103	0.0736

Model II

Table 3: LN_PRICE model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic
19748	19718	29	19747	30	0.53225	0.37275	0.37183	0.53185	0.42924	404.0598	0	31166

Table 4: LN_PRICE model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.9436	566.7317	0.0193	0.0000	10.9057	10.9814
ln_landratio	0.2395	13.4905	0.0178	0.0000	0.2047	0.2743
ln_base_bldgsize_ratio	0.8885	22.0308	0.0403	0.0000	0.8095	0.9676
single_family1story	0.0400	3.6384	0.0110	0.0003	0.0184	0.0615
half_duplex1story	-0.1507	-3.1963	0.0471	0.0014	-0.2430	-0.0583
half_duplex2story	-0.1788	-3.9366	0.0454	0.0001	-0.2678	-0.0898
smallest	0.0854	3.7761	0.0226	0.0002	0.0411	0.1297
small	0.0311	2.2710	0.0137	0.0232	0.0043	0.0580
two_plus_baths	0.1567	5.2032	0.0301	0.0000	0.0977	0.2158
powder_room	0.0841	5.2959	0.0159	0.0000	0.0530	0.1153
good	0.3456	11.7740	0.0293	0.0000	0.2880	0.4031
fair	-0.2042	-21.5599	0.0095	0.0000	-0.2228	-0.1856
poor	-0.3548	-16.3833	0.0217	0.0000	-0.3972	-0.3123
very_poor_unsound	-0.3308	-7.3216	0.0452	0.0000	-0.4193	-0.2422
qabove_avg	0.4114	14.3730	0.0286	0.0000	0.3553	0.4675
qbelow_avg	-0.0408	-4.1161	0.0099	0.0000	-0.0603	-0.0214
yb_pre_1929	-0.1572	-12.9904	0.0121	0.0000	-0.1809	-0.1335
yb_1946_to_1960	0.0599	6.4402	0.0093	0.0000	0.0417	0.0782
slab	-0.0542	-1.7769	0.0305	0.0756	-0.1140	0.0056
ln_garagespaces	0.0842	8.4115	0.0100	0.0000	0.0646	0.1038
fireplace	0.0996	10.5298	0.0095	0.0000	0.0811	0.1182
hot_water	0.0701	4.7234	0.0148	0.0000	0.0410	0.0991
elec_baseboard	-0.3420	-2.3882	0.1432	0.0169	-0.6227	-0.0613
wall	-0.1246	-2.5749	0.0484	0.0100	-0.2194	-0.0297

term	estimate	statistic	std_error	p_value	conf_low	conf_high
central_air	0.0906	8.4656	0.0107	0.0000	0.0697	0.1116
months_1to9	-0.0100	-4.8231	0.0021	0.0000	-0.0141	-0.0059
months_10to16	0.0188	8.5808	0.0022	0.0000	0.0145	0.0231
months_17to24	0.0102	10.7513	0.0009	0.0000	0.0083	0.0121
near_downtown_1mi	1.4331	5.3451	0.2681	0.0000	0.9076	1.9587
near_downtown_5mi	0.0637	3.9387	0.0162	0.0001	0.0320	0.0954

Model III

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	within_10_pct	within_20_pct	within_50_pct
19748	0.9859799	1.154115	0.8770936	1.31584	48.28461	-0.7462241	13.7381	27.95726	68.2854

Table 6: LN_PRICE (base) model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic
19748	19718	29	19747	30	0.53225	0.37275	0.37183	0.53185	0.42924	404.0598	0	31166 31

Table 7: LN_PRICE (+ECFs) model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.9436	566.7317	0.0193	0.0000	10.9057	10.9814
ln_landratio	0.2395	13.4905	0.0178	0.0000	0.2047	0.2743
ln_base_bldgsize_ratio	0.8885	22.0308	0.0403	0.0000	0.8095	0.9676
single_family1story	0.0400	3.6384	0.0110	0.0003	0.0184	0.0615
half_duplex1story	-0.1507	-3.1963	0.0471	0.0014	-0.2430	-0.0583
half_duplex2story	-0.1788	-3.9366	0.0454	0.0001	-0.2678	-0.0898
smallest	0.0854	3.7761	0.0226	0.0002	0.0411	0.1297
small	0.0311	2.2710	0.0137	0.0232	0.0043	0.0580
two_plus_baths	0.1567	5.2032	0.0301	0.0000	0.0977	0.2158
powder_room	0.0841	5.2959	0.0159	0.0000	0.0530	0.1153
good	0.3456	11.7740	0.0293	0.0000	0.2880	0.4031
fair	-0.2042	-21.5599	0.0095	0.0000	-0.2228	-0.1856
poor	-0.3548	-16.3833	0.0217	0.0000	-0.3972	-0.3123
very_poor_unsound	-0.3308	-7.3216	0.0452	0.0000	-0.4193	-0.2422
qabove_avg	0.4114	14.3730	0.0286	0.0000	0.3553	0.4675
qbelow_avg	-0.0408	-4.1161	0.0099	0.0000	-0.0603	-0.0214
yb_pre_1929	-0.1572	-12.9904	0.0121	0.0000	-0.1809	-0.1335
yb_1946_to_1960	0.0599	6.4402	0.0093	0.0000	0.0417	0.0782
slab	-0.0542	-1.7769	0.0305	0.0756	-0.1140	0.0056
ln_garagespaces	0.0842	8.4115	0.0100	0.0000	0.0646	0.1038
fireplace	0.0996	10.5298	0.0095	0.0000	0.0811	0.1182
hot_water	0.0701	4.7234	0.0148	0.0000	0.0410	0.0991
elec_baseboard	-0.3420	-2.3882	0.1432	0.0169	-0.6227	-0.0613
wall	-0.1246	-2.5749	0.0484	0.0100	-0.2194	-0.0297
central_air	0.0906	8.4656	0.0107	0.0000	0.0697	0.1116
months_1to9	-0.0100	-4.8231	0.0021	0.0000	-0.0141	-0.0059
months_10to16	0.0188	8.5808	0.0022	0.0000	0.0145	0.0231
months_17to24	0.0102	10.7513	0.0009	0.0000	0.0083	0.0121

term	estimate	statistic	std_error	p_value	conf_low	conf_high
near_downtown_1mi	1.4331	5.3451	0.2681	0.0000	0.9076	1.9587
near_downtown_5mi	0.0637	3.9387	0.0162	0.0001	0.0320	0.0954

Table 8: VIF / GVIF

term	vif	tolerance
ln_base_bldgsize_ratio	9.604	0.104
months_10to16	3.242	0.308
smallest	3.194	0.313
two_plus_baths	2.821	0.355
small	2.460	0.407
powder_room	2.229	0.449
months_1to9	2.002	0.499
months_17to24	1.998	0.500
yb_pre_1929	1.952	0.512
qbelow_avg	1.683	0.594
single_family1story	1.654	0.605
qabove_avg	1.491	0.671
fireplace	1.468	0.681
hot_water	1.468	0.681
yb_1946_to_1960	1.381	0.724
near_downtown_5mi	1.375	0.727
ln_landratio	1.322	0.757
ln_garagespaces	1.129	0.886
central_air	1.092	0.916
half_duplex2story	1.091	0.917
half_duplex1story	1.090	0.917
fair	1.058	0.945
wall	1.043	0.959
slab	1.041	0.960
poor	1.040	0.961
good	1.031	0.970
very_poor_unsound	1.016	0.984
near_downtown_1mi	1.015	0.985
elec_baseboard	1.013	0.987

Model IV

Table 9: LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	within_10_pct	within_20_pct	within_50_pct
18421	1.164941	1.308701	1.063421	1.230651	41.86557	-0.7512901	14.39661	29.23294	66.04962

Table 10: Alternative LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	within_10_pct	within_20_pct	within_50_pct
18421	1.164344	1.308737	1.061096	1.233383	41.89053	-0.7517307	14.4129	29.56408	66.00619

Table 11: LN_TASP model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	b
18421	18396	24	18420	26	0.45831	0.43932	0.43859	0.458	0.38096	600.5914	0	23559	2376

Table 12: LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.1752	808.8518	0.0138	0.0000	11.1481	11.2022
ln_landratio	0.2869	17.7032	0.0162	0.0000	0.2551	0.3186
ln_base_bldgsize_ratio	0.9297	24.9960	0.0372	0.0000	0.8568	1.0026
single_family1story	0.0232	2.4063	0.0097	0.0161	0.0043	0.0422
half_duplex1story	-0.2143	-4.1231	0.0520	0.0000	-0.3162	-0.1124
half_duplex2story	-0.2640	-5.5190	0.0478	0.0000	-0.3578	-0.1703
smallest	0.0978	4.7861	0.0204	0.0000	0.0578	0.1379
small	0.0385	3.1461	0.0122	0.0017	0.0145	0.0625
two_plus_baths	0.1410	4.9572	0.0284	0.0000	0.0852	0.1967
powder_room	0.0855	5.9082	0.0145	0.0000	0.0571	0.1139
good	0.3734	12.6387	0.0295	0.0000	0.3155	0.4314
fair	-0.2050	-24.2668	0.0084	0.0000	-0.2215	-0.1884
poor	-0.3817	-17.6954	0.0216	0.0000	-0.4239	-0.3394
very_poor_unsound	-0.3563	-5.9514	0.0599	0.0000	-0.4736	-0.2390
qabove_avg	0.4037	15.1326	0.0267	0.0000	0.3514	0.4560
qbelow_avg	-0.0474	-5.3754	0.0088	0.0000	-0.0647	-0.0301
yb_pre_1929	-0.1613	-14.6916	0.0110	0.0000	-0.1828	-0.1398
yb_1946_to_1960	0.0607	7.4164	0.0082	0.0000	0.0447	0.0767
crawl	-0.0560	-2.3311	0.0240	0.0198	-0.1032	-0.0089
slab	-0.0577	-1.7829	0.0323	0.0746	-0.1210	0.0057
ln_garagespaces	0.0833	9.2969	0.0090	0.0000	0.0658	0.1009
fireplace	0.1048	12.3806	0.0085	0.0000	0.0882	0.1214
hot_water	0.0763	5.5781	0.0137	0.0000	0.0495	0.1031
central_air	0.0950	10.1154	0.0094	0.0000	0.0766	0.1134
near_downtown_1mi	NA	NA	NA	NA	NA	NA
near_downtown_5mi	0.0296	1.8951	0.0156	0.0581	-0.0010	0.0602

Table 13: Alternative LN_TASP model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	b
18421	18396	24	18420	26	0.45853	0.43878	0.43805	0.45822	0.38092	599.2749	0	23577	2376

Table 14: Alternative LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	7.9312	41.7160	0.1901	0.0000	7.5585	8.3038
ln_landratio	0.3067	19.1069	0.0160	0.0000	0.2752	0.3381
lnsbldsfint	0.5032	17.3578	0.0290	0.0000	0.4464	0.5601
arterial	-0.0212	-1.6573	0.0128	0.0975	-0.0463	0.0039
single_family1story	0.0259	2.7294	0.0095	0.0064	0.0073	0.0445
smallest	0.1216	4.7049	0.0258	0.0000	0.0709	0.1722
small	0.0312	2.3862	0.0131	0.0170	0.0056	0.0567
large	0.0407	2.9623	0.0138	0.0031	0.0138	0.0677
largest	0.0740	2.9848	0.0248	0.0028	0.0254	0.1226
two_plus_baths	0.2367	8.1565	0.0290	0.0000	0.1798	0.2936
powder_room	0.1082	7.1303	0.0152	0.0000	0.0784	0.1379
good	0.3747	12.6758	0.0296	0.0000	0.3168	0.4326
fair	-0.2029	-24.0062	0.0085	0.0000	-0.2195	-0.1863
poor	-0.3796	-17.5899	0.0216	0.0000	-0.4219	-0.3373

term	estimate	statistic	std_error	p_value	conf_low	conf_high
qbelow_avg	-0.0468	-5.2954	0.0088	0.0000	-0.0641	-0.0295
yb_pre_1929	-0.1565	-14.2402	0.0110	0.0000	-0.1781	-0.1350
yb_1946_to_1960	0.0629	7.6700	0.0082	0.0000	0.0468	0.0790
crawl	-0.0451	-1.8735	0.0241	0.0610	-0.0923	0.0021
garage	0.0694	9.0889	0.0076	0.0000	0.0544	0.0843
fireplace	0.1039	12.2719	0.0085	0.0000	0.0873	0.1205
hot_water	0.0737	5.3656	0.0137	0.0000	0.0468	0.1007
central_air	0.0972	10.3603	0.0094	0.0000	0.0788	0.1156
near_downtown_1mi	NA	NA	NA	NA	NA	NA
near_downtown_5mi	0.0362	2.3112	0.0157	0.0208	0.0055	0.0669

Table 15: VIF / GVIF

term	vif	tolerance
ln_base_bldgsize_ratio	9.811	0.102
smallest	3.248	0.308
two_plus_baths	2.806	0.356
small	2.486	0.402
powder_room	2.267	0.441
yb_pre_1929	1.899	0.527
qbelow_avg	1.678	0.596
single_family1story	1.553	0.644
qabove_avg	1.536	0.651
fireplace	1.487	0.672
hot_water	1.473	0.679
yb_1946_to_1960	1.366	0.732
near_downtown_5mi	1.330	0.752
ln_landratio	1.303	0.767
ln_garagespaces	1.120	0.893
central_air	1.086	0.920
half_duplex2story	1.062	0.941
half_duplex1story	1.050	0.952
fair	1.047	0.955
crawl	1.040	0.962
poor	1.025	0.976
good	1.017	0.983
slab	1.009	0.991
very_poor_unsound	1.004	0.996

Table 16: VIF / GVIF

term	vif	tolerance
lnsbldsfint	15.537	0.064
largest	5.289	0.189
smallest	5.183	0.193
large	3.032	0.330
two_plus_baths	2.918	0.343
small	2.824	0.354
powder_room	2.489	0.402
yb_pre_1929	1.902	0.526
qbelow_avg	1.683	0.594
qabove_avg	1.537	0.650
single_family1story	1.497	0.668
hot_water	1.485	0.673

term	vif	tolerance
fireplace	1.484	0.674
yb_1946_to_1960	1.370	0.730
near_downtown_5mi	1.338	0.747
ln_landratio	1.277	0.783
garage	1.096	0.912
central_air	1.084	0.923
fair	1.047	0.955
crawl	1.041	0.961
poor	1.025	0.975
arterial	1.019	0.982
good	1.017	0.983
very_poor_unsound	1.003	0.997

